DTU Compute
Department of Applied Mathematics and Computer Science

Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



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Estimating Thermal Comfort Curves with Hidden Markov Models

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7 December, 2026

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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

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Signature

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Date

Abstract

ABSTRACT IS MISSING

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Mads Holme Håkonsson, MSc Civil Engineering, DTU
Creator of this thesis template.

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1 Introduction

Reduction in the usage of energy in buildings is a crucial aspect if one wants to go towards a more sustainable future. It has long been known that buildings are responseible for X% of the total energy consumption in the world, and that a large part of this energy is for HVAC systems [MISSING REFERENCE] HVAC systems energy consumption has been directly related to the occupant behaviours of the buildings. This imposes a challenge, as one has to be able to understand the behaviour of the occupants in the buildings before we can gain the largest optimization in energy consumption. (MISSING REFERENCE) has shown that there is a large variability in the behavour of occupants such as thermal comfort, lighting & ventiliation needs and precense hours.

Within the context of energy consumption in buildings, heating has been estimated to contribute to X% [MISSING REFERENCE] of the consumption. An aspect of heating of a building is the byproduct of the thermal comfort of the occupants in the building, and therefore, if we can understand the thermal comforts of the occupants in the buildings, we can optimize the energy consumption drasitically. However, the thermal comfort curves is a personal preference of the occupants, and we should therefore expect this to differer dependent on the individual buildings and the number of occupants.

This projects aims to understand when the occupants of a single offices are present, and how much thermal comfort feedback does we need to collect from the occupants in order to be able to infer the true thermal comfort of the occupants in the building. The goal is to also build a codebase others and easyly download and extent.
We will do this by using a Hidden Markov Model (HMM) to model the state of the room

WRITTEN OUTLINE OF EACH CHAPTER AND WHAT QUESTIONS THEY ANSWER

2 Data

The data used in this project are CO_2 data collected from the office building 303B and the censors and data providers are climify [MISSING REFERENCE]. The data is collected in the fall and winter period of 2023, and the raw data period spans from [MISSING REFERENCE] to [MISSING REFERENCE]. The data collected from the censors are in intervals of around 10 minutes. Because our HMM models works in discrete time, the data have been aggregated into 30 minutes such that each time step $\Delta t = 1$ represents 30 minutes.

When collecting CO_2 data, the censors tend to drift over time [MISSING REFERENCE], and therefore, the data collected from the censors are not always accurate. To empty rooms, that have the same CO_2 levels may show different readins based on how much the censors have drifted over time. This is a problem, as we want to use the CO_2 data to infer the presence of the occupants in the building and a higher baseline of the CO_2 data may lead to a false positive of the presence of the occupants in the building. To correct for this, we have used a method called *Automatic Baseline Correction* (ABC) Algorithm. [MISSING REFERENCE]. In the use of this algorithm, we have assumed that the lowest CO_2 ppm values over a period is equal to the outside CO_2 ppm values. From literature, it's common to set the outside CO_2 ppm values to 400 ppm [MISSING REFERENCE], which we have also done.

Let C_t be the CO_2 ppm value at time t, $\tilde{C}_t$ be the corrected CO_2 ppm value at time t.

We then define the modified ABC algorithm as

$$\tilde{C}_t = C_t - Q_{0.02}\left(C_{t-\Delta t:t}\right) + 400, \tag{2.1}$$

where $Q_{0.02}\left(C_{t-\Delta t:t}\right)$ is the 2% quantile of the CO_2 ppm values over the time period $[t - \Delta t, t]$ and Δt is a fixed time period. In this project, we have set $\Delta t = 8$ days, such that the weekday overlaps. The reason for using a 2% quantile is to avoid the effect of erradic readings of the CO_2 ppm values, which may occur if the censors are not working properly. Because we are interrested in predicting future presence of the o

A Title

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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